## Exercise C - II

1. Solve the equation : $\cos^{-1}\left(\sqrt{6}x\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2}$

हल कीजिए : $\cos^{-1}\left(\sqrt{6}x\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2}$

**Ans.** $x = 1/3$

**Sol.** $\cos^{-1}\left(\sqrt{6}x\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2}$

$\sin^{-1}\left(\sqrt{1-6x^2}\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2}$

$\sqrt{1-6x}=3\sqrt{3}x^2$

$1 - 6x^2 = 27x^4$

$27x^4 + 6x^2 - 1 = 0$

$27x^4 + 9x^2 - 3x^2 - 1 = 0$

$9x^2\ (3x^2 + 1) - 1\ (3x^2 + 1) = 0$

$x^2=\frac{1}{9}\Rightarrow x=\pm\frac{1}{3}$ $\qquad$ $x^2=-\frac{1}{3}$ (not possible)

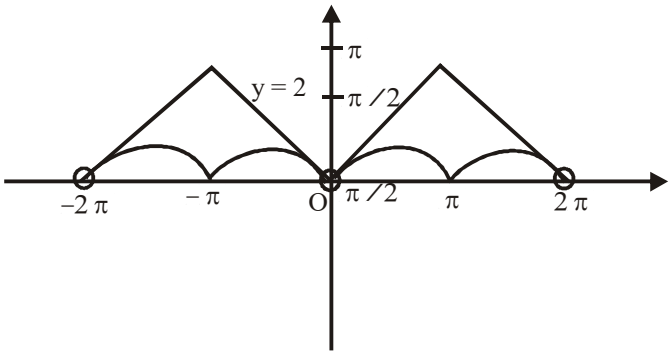
$x=\frac{1}{3}$, $x > 0$

2. Find the number of ordered pairs of (x, y), which satisfy $y = |\sin x|$ and $y = \cos^{-1}(\cos x)$, where $-2\pi \le x \le 2\pi$.

अन्तराल $-2\pi \le x \le 2\pi$ उन क्रमित युग्मों की संख्या ज्ञात करो जो $y = |\sin x|$ तथा $y = \cos^{-1}(\cos x)$ को सन्तुष्ट कर सके।

**Ans.** 3

**Sol.** $y = |\sin x|$ ; $y = \cos^{-1}(\cos x)$



3. Prove that $\cos \tan^{-1} \sin \cot^{-1} x = \sqrt{\frac{x^2+1}{x^2+2}}$.

सिद्ध कीजिए $\cos \tan^{-1} \sin \cot^{-1} x = \sqrt{\frac{x^2+1}{x^2+2}}$

**Ans.**

**Sol.** $\text{LHS} = \cos \tan^{-1} \sin \sin^{-1} \left( \frac{1}{\sqrt{1+x^2}} \right)$

$= \cos \tan^{-1} \left( \frac{1}{\sqrt{1+x^2}} \right)$

$= \cos \cos^{-1} \left( \frac{\sqrt{x^2+1}}{\sqrt{2+x^2}} \right)$

$= \sqrt{\frac{x^2+1}{x^2+2}}$

4. Prove that, $\tan^{-1} \left( \frac{1}{2} \tan 2A \right) + \tan^{-1} (\cot A) + \tan^{-1} (\cot^3 A) = \begin{cases} \pi & , \text{ if } 0 < A < \frac{\pi}{4} \\ 0 & , \text{ if } \frac{\pi}{4} < A < \frac{\pi}{2} \end{cases}$.

सिद्ध कीजिए $\tan^{-1} \left( \frac{1}{2} \tan 2A \right) + \tan^{-1} (\cot A) + \tan^{-1} (\cot^3 A) = \begin{cases} \pi & , \text{ if } 0 < A < \frac{\pi}{4} \\ 0 & , \text{ if } \frac{\pi}{4} < A < \frac{\pi}{2} \end{cases}$

**Ans.**

**Sol.** $\text{LHS} = \tan^{-1} \left( \frac{1}{2} \tan 2A \right) + \tan^{-1} \left( \cot A \right) + \tan^{-1} \left( \cot^3 A \right)$

Case – I$^{st}$ : $0 < A < \frac{\pi}{4}$

$(\cot A)\,(\cot^3 A) > 1$

$\tan^{-1} \left( \frac{1}{2} \tan 2A \right) + \pi + \tan^{-1} \left( \frac{\cot A + \cot^3 A}{1 - \cot^4 A} \right)$

$$= \tan^{-1}\left(\frac{1}{2}\tan 2A\right) + \pi + \tan^{-1}\left(\frac{\cot A\left(1+\cot^2 A\right)}{\left(1-\cot^2 A\right)\left(1+\cot^2 A\right)}\right)$$

$$= \pi + \tan^{-1}\left(\frac{1}{2}\tan 2A\right) + \tan^{-1}\left(\frac{\frac{1}{\tan A}}{\frac{\tan^2 A - 1}{\tan^2 A \tan A}}\right)$$

$$= \pi + \tan^{-1}\left(\frac{1}{2}\tan 2A\right) + \tan^{-1}\left(\frac{1}{2}\times\frac{-2\tan A}{1-\tan^2 A}\right)$$

$$= \pi + \tan^{-1}\left(\frac{1}{2}\tan 2A\right) - \tan^{-1}\left(\frac{1}{2}\tan 2A\right)$$

$= \pi$

Case – II[nd] : $\frac{\pi}{4} < A < \frac{\pi}{2}$

$$\Rightarrow \tan^{-1}\left(\frac{1}{2}\tan 2A\right) - \tan^{-1}\left(\frac{1}{2}\tan 2A\right)$$

$= 0$

5. Let $f(x) = \frac{\pi}{4} + \cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) - \tan^{-1} x$ and $a_i \left(a_i < a_{i+1} \ \forall\ i = 1,2,3,.....,n\right)$ be the positive integral values of x for which $\operatorname{sgn}\left(f(x)\right) = 1$, where sgn(y) denotes signum function of y. Find $\sum_{i=1}^{n} a_i^2$.

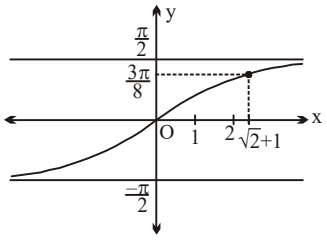
माना $f(x) = \frac{\pi}{4} + \cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) - \tan^{-1} x$ तथा $a_i \left(a_i < a_{i+1} \ \forall\ i = 1,2,3,.....,n\right)$, x के सभी धनात्मक पूर्णांक मान है जिसके लिये $\operatorname{sgn}\left(f(x)\right) = 1$, जहाँ sgn(y), y का सिग्नम फलन है। $\sum_{i=1}^{n} a_i^2$ ज्ञात कीजिये।

**Ans.** 5

**Sol.** We have $f(x)=\frac{\pi}{4}+\cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)-\tan^{-1}x = \frac{\pi}{4}+\cot^{-1}x-\tan^{-1}x = \frac{3\pi}{4}-2\tan^{-1}x$

Now, $\text{sgn}(f(x))=1 \Rightarrow f(x)>0$

So, $\frac{3\pi}{4}-2\tan^{-1}x>0 \Rightarrow \tan^{-1}x<\frac{3\pi}{8} \Rightarrow x<\sqrt{2}+1.$




$\therefore$ Possible positive integral values of $x$ are 1, 2.

So, $a_1=1$ and $a_2=2$.

Hence $\sum_{i=1}^{n} a_i^2 = \sum_{i=1}^{2} a_i^2 = a_1^2+a_2^2 = 1+4 = 5$ Ans.

6. If $\tan\left(\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+.....+\tan^{-1}\left(\frac{1}{381}\right)\right)=\frac{m}{n}$. where $m, n \in N$, then find least value of $(m+n)$.

यदि $\tan\left(\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+.....+\tan^{-1}\left(\frac{1}{381}\right)\right)=\frac{m}{n}$. जहाँ $m, n \in N$, तो $(m+n)$ का न्यूनतम मान होगा।

**Ans.** 40

**Sol.** We have $\tan\left(\tan^{-1}\left(\frac{1}{1+1\cdot2}\right)+\tan^{-1}\left(\frac{1}{1+2\cdot3}\right)+.....+\tan^{-1}\left(\frac{1}{1+19\cdot20}\right)\right)$

$=\tan\left(\tan^{-1}\left(\frac{2-1}{1+2\cdot1}\right)+\tan^{-1}\left(\frac{3-2}{1+2\cdot3}\right)+.....+\tan^{-1}\left(\frac{20-19}{1+19\cdot20}\right)\right)$

$=\tan[\tan^{-1}2-\tan1+\tan^{-1}3-\tan^{-1}2+.....+\tan^{-1}20-\tan^{-1}19] = \tan(\tan^{-1}20-\tan^{-1}1)$

$=\tan\left(\tan^{-1}\left(\frac{20-1}{1+20\cdot1}\right)\right)=\frac{20-1}{1+20}=\frac{19}{21}=\frac{m}{n}$ (Given)

$\therefore$ $m=19, n=21$

Hence $(m+n)_{\text{Least}} = 40$. Ans.

7. If $\sum_{n=0}^{\infty} 2\ \text{arc}\cot\left(\frac{n^2+n+4}{2}\right) = k\pi$, then find the value of k.

यदि $\sum_{n=0}^{\infty} 2\ \text{arc}\cot\left(\frac{n^2+n+4}{2}\right) = k\pi$, तो k का मान ज्ञात कीजिए।

**Ans.** 1

**Sol.** $\sum_{n=0}^{\infty} 2\cot^{-1}\left(\frac{n^2+n+4}{2}\right) = k\pi$

$$T_n = \tan^{-1}\left(\frac{2}{n^2+n+4}\right)$$

$$= \tan^{-1}\left(\frac{2}{4\left(1+\frac{n(n+1)}{4}\right)}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{1}{2}}{1+\frac{n}{2}\left(\frac{n+1}{2}\right)}\right)$$

$$T_n = \tan^{-1}\left(\frac{n+1}{2}\right) - \tan^{-1}\left(\frac{n}{2}\right)$$

$$T_0 = \tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}(0)$$

$$T_1 = \tan^{-1}\left(\frac{2}{2}\right) - \tan^{-1}\left(\frac{1}{2}\right)$$

$$T_2 = \tan^{-1}\left(\frac{3}{2}\right) - \tan^{-1}\left(\frac{2}{2}\right)$$

$\vdots \qquad \vdots \qquad \vdots$

$$T_n = \tan^{-1}\left(\frac{n+1}{2}\right) - \tan^{-1}\left(\frac{n}{2}\right)$$

$$S_n = \tan^{-1}\left(\frac{n+1}{2}\right) - \tan^{-1}(0)$$

$$S_\infty = \frac{\pi}{2} - 0$$

$$2S_\infty = \pi = k\pi$$

$k = 1$

8. Let $\alpha, \beta$ and $\gamma$ are the roots of the equation $x^3 + 6x + 3 = 0$ and

$$A = \cos^{-1}\left(\sin\left((\alpha+\beta)^{-1} + (\beta+\gamma)^{-1} + (\gamma+\alpha)^{-1}\right)\right),\ B = \cos\left(\tan^{-1}\left(\sin\left(\frac{\alpha+\beta+\gamma}{2}\right)\right)\right)\ C = \sec^{-1}$$

$$\left(\operatorname{cosec}\left((1-\alpha)(1-\beta)(1-\gamma)\right)\right).$$

(i) If the range of the quadratic trinomial $g(x) = x^2 - 2Bx + k$ is $[0, \infty)$, then find the range of k.

(ii) Find the value of $(5A + B - C)$.

(iii) Find the range of the function $f(x) = \dfrac{(5A-C)x^5 + 6Bx^2}{x^4 + (B-1)x^3 + 1}$.

यदि $\alpha, \beta$ तथा $\gamma$ समीकरण $x^3 + 6x + 3 = 0$ के मूल है और

$$A = \cos^{-1}\left(\sin\left((\alpha+\beta)^{-1} + (\beta+\gamma)^{-1} + (\gamma+\alpha)^{-1}\right)\right),\ B = \cos\left(\tan^{-1}\left(\sin\left(\frac{\alpha+\beta+\gamma}{2}\right)\right)\right)\ C = \sec^{-1}$$

$$\left(\operatorname{cosec}\left((1-\alpha)(1-\beta)(1-\gamma)\right)\right).$$

(i) यदि द्विघात व्यंजक $g(x) = x^2 - 2Bx + k$ का परिसर $[0, \infty)$ हो, तो k का परिसर ज्ञात कीजिए।

(ii) $(5A + B - C)$ का मान ज्ञात कीजिए।

(iii) फलन $f(x) = \dfrac{(5A-C)x^5 + 6Bx^2}{x^4 + (B-1)x^3 + 1}$ का परिसर ज्ञात कीजिए।

**Ans.** (i) $\{1\}$ (ii) 1 (iii) $[0, 3]$

**Sol.** $x^3 + 6x + 3 = 0$

$\alpha + \beta + \gamma = 0$ $\alpha\beta + \beta\gamma + \gamma\alpha = 6$ $\alpha\beta\gamma = -3$

$A = \cos^{-1}(\sin(\alpha+\beta)^{-1} + (\alpha+\beta)^{-1} + (\gamma+\alpha)^{-1})$

$$= \cos^{-1}\left[\sin\left(-\left(\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}\right)\right)\right]$$

$$= \cos^{-1}\left[\sin\left(\frac{-(\alpha\beta+\beta\gamma+\gamma\alpha)}{\alpha\beta\gamma}\right)\right]$$

$$= \cos^{-1}\left[\sin\left(\frac{6}{3}\right)\right]$$

$= \cos^{-1}[\sin(2)]$

$$= \cos^{-1}\left[\cos\left(\frac{\pi}{2}-2\right)\right]$$

$$A \Rightarrow 2-\frac{\pi}{2}$$

$$B = \cos\left[\tan^{-1}\left(\sin\left(\frac{\alpha+\beta+\gamma}{2}\right)\right)\right]$$

$= \cos[\tan^{-1}(\sin(0))]$

$= \cos[\tan^{-1}(0)]$

$= \cos(0)$

$B = 1$

$C = \sec^{-1}(\text{cosec}((1-\alpha)(1-\beta)(1-\gamma)))$

$x^3 + 6x + 3 = (x-\alpha)(x-\beta)(x-\gamma)$

$x = 1$

$1 + 6 + 3 = (1-\alpha)(1-\beta)(1-\gamma)$

$(1-\alpha)((1-\beta)(1-\gamma) = 10$

$C = \sec^{-1}(\text{cosec } 10)$

$$= \sec^{-1}\left(\sec\left(10-\frac{\pi}{2}\right)\right)$$

$$= 10-\frac{\pi}{2}-2\pi$$

$$C = 10-\frac{5\pi}{2}$$

(i) $g(x) = x^2 - 2Bx + k$

$B = 1$

$= x^2 - 2x + k$

$$-\frac{D}{4a} = 0$$

$$\frac{-(4-4k)}{4} = 0$$

$k - 1 = 0$

$k = 1$

(ii) $5A + B - C = 5\left(2-\frac{\pi}{2}\right)+1-\left(10-\frac{5\pi}{2}\right)$

$$=10-\frac{5\pi}{2}+1-10+\frac{5\pi}{2}$$

$= 1$

(iii) $f(x)=\frac{(5A-C)x^5+6Bx^2}{x^4+(B-1)x^3+1}$

$$=\frac{\left(10-\frac{5\pi}{2}-10+\frac{5\pi}{2}\right)x^5+6x^2}{x^4+(1-1)x^3+1}$$

$y=\frac{6x^2}{x^4+1}\geq 0$ .....(1)

$yx^4 + y = 6x^2$

$yx^4 - 6x^2 + y = 0$

Case – I : $y \neq 0$

$D \geq 0$

$36 - 4y^2 \geq 0$

$+ 4 (y^2 - 9) \leq 0$

$(y - 3) (y + 3) \leq 0$

$y \in [-3,3] - \{0\}$ .....(2)

Case – II : $y = 0$

$0 \leq \frac{6x^2}{x^4+1}$

$x = 0$ .....(3)

$(2) \cup (3)$

$x = [-3, 3]$ .....(4)

$(1) \cap (4)$

$y = [0, 3]$

9. If $x \in \left[-1,-\frac{1}{2}\right]$ then express the function $f(x) = \sin^{-1}(3x - 4x^3) + \cos^{-1}(4x^3 - 3x)$ in the form of $a\cos^{-1} x + b\pi$, where a and b are rational numbers.

यदि $x \in \left[-1,-\frac{1}{2}\right]$ तो $f(x) = \sin^{-1}(3x - 4x^3) + \cos^{-1}(4x^3 - 3x)$ को $a\cos^{-1} x + b\pi$ के रूप में प्रदर्शित कीजिए, जहाँ a व b परिमेय संख्याएँ है।

**Ans.** $6\cos^{-1} x - \frac{9\pi}{2}$, so $a = 6, b = -\frac{9}{2}$

**Sol.** $f(x) = \sin^{-1}(3x - 4x^3) + \cos^{-1}(4x^3 - 3x)$

$\Rightarrow -\sin^{-1}(4x^3 - 3x) + \cos^{-1}(4x^3 - 3x)$

$$\Rightarrow \left(-\frac{\pi}{2} + \cos^{-1}\left(4x^3 - 3x\right)\right) + \cos^{-1}\left(4x^3 - 3x\right)$$

$$= -\frac{\pi}{2} + 2\cos^{-1}\left(4x^3 - 3x\right)$$

$\theta = \cos^{-1} x \qquad \theta \in \left[\frac{2\pi}{3}, \pi\right]$

$$= -\frac{\pi}{2} + 2\cos^{-1}\left(\cos 3\theta\right) \qquad 3\theta \in [2\pi, 3\pi]$$

$$= -\frac{\pi}{2} + 2\left(3\theta - 2\pi\right)$$

$$= -\frac{\pi}{2} + 6\theta - 4\pi$$

$$\Rightarrow 6\theta - \frac{9\pi}{2}$$

$$\Rightarrow 6\cos^{-1} x - \frac{9\pi}{2}$$

10. If $\alpha$ and $\beta$ are the roots of the equation $x^2 - 4x + 1 = 0$ $(\alpha > \beta)$ then find the value of

$$f(\alpha, \beta) = \frac{\beta^3}{2}\operatorname{cosec}^2\left(\frac{1}{2}\tan^{-1}\frac{\beta}{\alpha}\right) + \frac{\alpha^3}{2}\sec^2\left(\frac{1}{2}\tan^{-1}\frac{\alpha}{\beta}\right).$$

यदि $\alpha$ व $\beta$ समीकरण $x^2 - 4x + 1 = 0$ $(\alpha > \beta)$ के मूल है तो $f(\alpha, \beta) = \frac{\beta^3}{2}\operatorname{cosec}^2\left(\frac{1}{2}\tan^{-1}\frac{\beta}{\alpha}\right) + \frac{\alpha^3}{2}\sec^2\left(\frac{1}{2}\tan^{-1}\frac{\alpha}{\beta}\right)$ का मान ज्ञात कीजिए।

**Ans.** 56

**Sol.** $x^2 - 4x + 1 = 0 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right.$

$\alpha + \beta = 4 \qquad \alpha\beta = 1$

$$x = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

$$\alpha = 2 + \sqrt{3}, \beta = 2 - \sqrt{3}$$

11. Find the integral values of K for which the system of equations; $\begin{bmatrix} \arccos x + (\arcsin y)^2 & = \frac{K\pi^2}{4} \\ (\arcsin y)^2 \cdot (\arccos x) & = \frac{\pi^4}{16} \end{bmatrix}$ possesses solutions and find those solutions.

K का पूर्णांक मान ज्ञात कीजिए जिसके लिये दिये गये समीकरणों का हल हो तथा वह हल भी ज्ञात कीजिए।

$$\begin{bmatrix} \arccos x + (\arcsin y)^2 & = \frac{K\pi^2}{4} \\ (\arcsin y)^2 \cdot (\arccos x) & = \frac{\pi^4}{16} \end{bmatrix}.$$

**Ans.** $k = 2, \left(\cos\frac{\pi^2}{4}, 1\right)$ and $\left(\cos\frac{\pi^2}{4}, -1\right)$

**Sol.** Let $\cos^{-1}x = p$, $\sin^{-1}y = q$

$$p + q^2 = \frac{k\pi^2}{4}$$

$$p \cdot q^2 = \frac{\pi^4}{16}$$

$$x^2 - \frac{k\pi^2}{4}x + \frac{\pi^4}{16} = 0 \begin{matrix} \nearrow p \\ \searrow q^2 \end{matrix}$$

$$x^2 - \frac{k\pi^2}{4}x + \left(\frac{\pi^2}{4}\right)^2 = 0$$

So $k = 2$

$$x = \frac{\pi^2}{4}$$

$$q^2 = \frac{\pi^2}{4}$$

$$\sin^{-1} y = \pm\frac{\pi}{2}$$

$$y = \sin\left(\pm\frac{\pi}{2}\right) = \pm 1$$

$$x = \cos\left(\frac{\pi^2}{4}\right)$$

12. Solve the following inequalities :

असमिकाओं को हल कीजिए :

(a) $\text{arc cot}^2 x - 5 \text{ arc cot} x + 6 > 0$ (b) $\text{arc sin} x > \text{arc cos} x$

(c) $\tan^2(\text{arc sin} x) > 1$

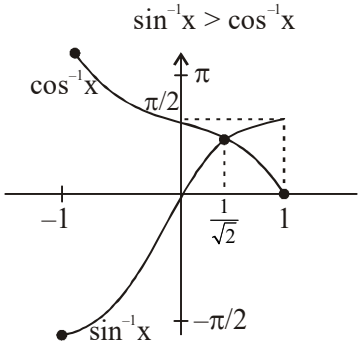
**Ans.** (a) $(-\infty, \cot 3) \cup (\cot 2, \infty)$ (b) $\left(\frac{\sqrt{2}}{2}, 1\right]$ (c) $\left(-1, -\frac{\sqrt{2}}{2}\right) \cup \left(\frac{\sqrt{2}}{2}, 1\right)$

**Sol.** (a) $\left(\cot^{-1} x - 2\right)\left(\cot^{-1} x - 3\right) > 0$

$$\cot^{-1} x \in (-\infty, 2) \cup (3, \infty)$$

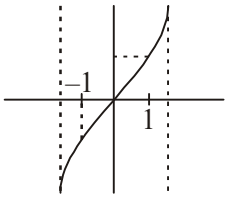
$$x \in (-\infty, \cot 3) \cup (\cot 2, \infty)$$

(b)




$$x \in \left(\frac{1}{\sqrt{2}}, 1\right]$$

(c)




$$\tan^2\left(\sin^{-1} x\right) > 1$$

Let $\theta$

$$\tan^2 \theta > 1$$

$$\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$x \in \left(-1, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$$

13. (a) Solve the inequality : $(\text{arc sec } x)^2 - 6(\text{arc sec } x) + 8 > 0$

(b) If $\sin^2 x + \sin^2 y < 1$ for all $x, y \in R$ then prove that $\sin^{-1}(\tan x \cdot \tan y) \in (-\pi/2, \pi/2)$.

(c) Solve the following system of inequations :
$4 \text{ arc } \tan^2 x - 8 \text{ arc } \tan x + 3 < 0$ and $4 \text{ arc cot } x - \text{arc cot}^2 x - 3 \geq 0$.

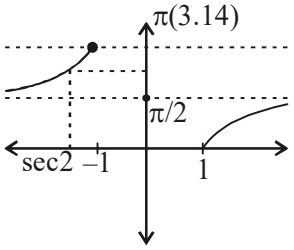
(a) दी गई असमिका को हल कीजिए : $(\text{arc sec } x)^2 - 6(\text{arc sec } x) + 8 > 0$

(b) यदि $\sin^2 x + \sin^2 y < 1 \ \forall \ x, y \in R$ तो सिद्ध कीजिये $\sin^{-1}(\tan x \cdot \tan y) \in \left(-\pi/2, \pi/2\right)$

(c) निम्न असमिकाओं के समूह का हल ज्ञात कीजिए :
$4 \text{ arc } \tan^2 x - 8 \text{ arc } \tan x + 3 < 0$ and $4 \text{ arc cot } x - \text{arc cot}^2 x - 3 \geq 0$.

**Ans.** (a) $(-\infty, \sec 2) \cup [1, \infty)$ (c) $\left(\tan\frac{1}{2}, \cot 1\right]$

**Sol.** (a) $\left(\sec^{-1} x - 2\right)\left(\sec^{-1} x - 4\right) > 0$




$\sec^{-1} x \in (-\infty, 2) \cup (4, \infty)$

$\sec^{-1} x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, 2\right)$

$x \in (-\infty, \sec 2) \cup [1, \infty)$

(b) $\sin^2 x < \cos^2 y$

$\Rightarrow \frac{\sin^2 x}{\cos^2 y} < 1$ _____(1)

Similarly,

$\frac{\sin^2 y}{\cos^2 x} < 1$ _____(2)

(1) × (2)

$\tan^2 x \cdot \tan^2 y < 1$

$-1 < \tan x \cdot \tan y < 1$

$-\frac{\pi}{2} < \sin^{-1}(\tan x \cdot \tan y) < \frac{\pi}{2}$

(c) $\left(2\tan^{-1} x - 1\right)\left(2\tan^{-1} x - 3\right) < 0$

$\tan^{-1} x \in \left(\frac{1}{2}, \frac{3}{2}\right)$

$x \in \left(\tan\frac{1}{2}, \tan\frac{3}{2}\right)$ _____(1)

$\left(\cot^{-1} x\right)^2 - 4\cot^{-1} x + 3 \le 0$

$\left(\cot^{-1} x - 1\right)\left(\cot^{-1} x - 3\right) \le 0$

$\cot^{-1} x \in [1, 3]$

$x \in [\cot 3, \cot 1]$ _____(2)

$(1) \cap (2)$

$\left(\tan\frac{1}{2}, \cot 1\right)$

**Paragraph for Question Nos. 14 to 16**

For $x \in \left(0, \frac{\pi}{4}\right)$, Let $S_n = \sum_{r=1}^{2n} \sin\left(\sin^{-1} x^{3r-2}\right)$, $C_n = \sum_{r=1}^{2n} \cos\left(\cos^{-1} x^{3r-1}\right)$ and $T_n = \sum_{r=1}^{2n} \tan\left(\tan^{-1} x^{3r}\right)$ where $n \in N$ and $n \geq 3$.

$x \in \left(0, \frac{\pi}{4}\right)$ के लिए, माना $S_n = \sum_{r=1}^{2n} \sin\left(\sin^{-1} x^{3r-2}\right)$, $C_n = \sum_{r=1}^{2n} \cos\left(\cos^{-1} x^{3r-1}\right)$ तथा $T_n = \sum_{r=1}^{2n} \tan\left(\tan^{-1} x^{3r}\right)$ जहाँ $n \in N$ एवं $n \geq 3$.

14. The correct order of $S_n$, $C_n$ and $T_n$ is given by :

$S_n$, $C_n$ एवं $T_n$ का सही क्रम होगा :

(A) $S_n > T_n > C_n$ (B) $S_n < C_n < T_n$ (C) $S_n < T_n < C_n$ (D) $S_n > C_n > T_n$

**Ans.** D

**Sol.** We have $S_n = x + x^4 + x^7 + x^{10} + \ldots\ldots$ (2n terms)

$C_n = x^2 + x^5 + x^8 + x^{11} + \ldots\ldots$ (2n terms)

$T_n = x^3 + x^6 + x^9 + x^{12} + \ldots\ldots$ (2n terms)

Clearly $S_n > C_n > T_n$ as x is a proper fraction $\therefore\ x > x^2 > x^3$ & so on

15. The value of $\lim_{n\to\infty} (S_n + C_n + T_n)$ is equal to :

$\lim_{n\to\infty} (S_n + C_n + T_n)$ का मान बराबर होगा :

(A) $\frac{1}{1-x}$ (B) $\frac{x}{1-x}$ (C) $\frac{1}{1-x}$ (D) $\frac{x}{1+x}$

**Ans.** B

**Sol.** We have $S_n = x + x^4 + x^7 + x^{10} + \ldots\ldots$ (2n terms)

$C_n = x^2 + x^5 + x^8 + x^{11} + \ldots\ldots$ (2n terms)

$T_n = x^3 + x^6 + x^9 + x^{12} + \ldots\ldots$ (2n terms)

$\lim\limits_{n\to\infty} (S_n + C_n + T_n) = x + x^2 + x^3 + \ldots\ldots \infty = \dfrac{x}{1-x}$

16. The value of 'x' for which $S_n = C_n + T_n$, is :

x का वह मान जिसके लिए $S_n = C_n + T_n$ हो, होगा :

(A) $\sin\dfrac{\pi}{5}$ (B) $2\sin\dfrac{\pi}{5}$ (C) $2\sin\dfrac{\pi}{10}$ (D) $\sin\dfrac{\pi}{10}$

**Ans.** C

**Sol.** We have $S_n = x + x^4 + x^7 + x^{10} + \ldots\ldots$ (2n terms)

$C_n = x^2 + x^5 + x^8 + x^{11} + \ldots\ldots$ (2n terms)

$T_n = x^3 + x^6 + x^9 + x^{12} + \ldots\ldots$ (2n terms)

We have $S_n = C_n + T_n \Rightarrow x\dfrac{\left((x^3)^{2n}-1\right)}{(x^3-1)} = x^2\dfrac{\left((x^3)^{2n}-1\right)}{(x^3-1)} + x^3\dfrac{\left((x^3)^{2n}-1\right)}{(x^3-1)}$

But $x \neq 1$, as $x \in \left(0, \dfrac{\pi}{4}\right)$, so we get $x = x^2 + x^3$

$\Rightarrow \quad x^2 + x - 1 = 0 \ (x \neq 0) \quad \Rightarrow \quad x = \dfrac{\sqrt{5}-1}{2} = 2\sin\dfrac{\pi}{10} \in \left(0, \dfrac{\pi}{4}\right)$

## MATCH THE COLUMN

17. Match the following columns :

| **Column-I** | **Column-II** |
|---|---|
| (A) $f(x) = \begin{cases} x\tan^{-1}\dfrac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ | (P) even |
| (B) $g(x) = \begin{cases} x\cot^{-1} x, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ | (Q) odd |
| (C) $h(x) = \begin{cases} x^2\tan^{-1}\dfrac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ | (R) neither even nor odd |

(D) $k(x) = \begin{cases} x\cot^{-1}x^2, & \text{if } x \neq 0 \\ 0 & , \text{ if } x = 0 \end{cases}$ (S) even as well as odd

निम्न कॉलम को सुमेलित कीजिए :

| **Column-I** | **Column-II** |
|---|---|
| (A) $f(x) = \begin{cases} x\tan^{-1}\frac{1}{x}, & \text{यदि } x \neq 0 \\ 0 & , \text{ यदि } x = 0 \end{cases}$ | (P) सम |
| (B) $g(x) = \begin{cases} x\cot^{-1}x, & \text{यदि } x \neq 0 \\ 0 & , \text{ यदि } x = 0 \end{cases}$ | (Q) विषम |
| (C) $h(x) = \begin{cases} x^2\tan^{-1}\frac{1}{x}, & \text{यदि } x \neq 0 \\ 0 & , \text{ यदि } x = 0 \end{cases}$ | (R) न तो सम न ही विषम |
| (D) $k(x) = \begin{cases} x\cot^{-1}x^2, & \text{यदि } x \neq 0 \\ 0 & , \text{ यदि } x = 0 \end{cases}$ | (S) सम तथा विषम दोनों |

**Ans.** (A) P (B) R (C) Q (D) Q

**Sol.** (A) $f(-x) = f(x) \Rightarrow f$ is even

(B) g(x) is neither even nor odd.

(C) $h(-x) = -h(x) \Rightarrow h(x)$ is odd.

(D) $k(-x) = -k(x) \Rightarrow k(x)$ is odd

18. Match the following column :

| **Column -I** | **Column -II** |
|---|---|
| (A) Let $f: R \to R$ be defined as $f(x) = \sqrt[3]{x} + \tan^{-1}x$, then f(x) is | (P) one-one |
| (B) Let $f: (-\infty, \infty) \to \{-1, 0, 1\}$ be defined as $f(x) = \sin^3\pi\left(\text{sgn}(x^2+3x+5)\right)$ then f(x) is (where sgn x denotes signum function of x) | (Q) into |
| (C) Let $f: [-2, 2] \to (0, e^2]$ be defined as $f(x) = e^{\sqrt{x^2}}$, then f(x) is | (R) odd |
| (D) Let $f: (-1, 5) \to [0, 3]$ be defined as $f(x) = \sqrt{5+4x-x^2}$, then f(x) is | (S) non-invertible |
| | (T) aperiodic |

निम्न कॉलम को सुमेलित कीजिए :

| **Column -I** | **Column -II** |
|---|---|
| (A) माना फलन $f: R \to R$ इस प्रकार की $f(x) = \sqrt[3]{x} + \tan^{-1} x$, तो f(x) होगा | (P) एकैकी |
| (B) माना फलन $f: (-\infty, \infty) \to \{-1, 0, 1\}$ इस प्रकार की $f(x) = \sin^3 \pi\left(\text{sgn}(x^2 + 3x + 5)\right)$ तो f(x) होगा (जहाँ sgn x, x का सिग्नम फलन है.) | (Q) अन्तःक्षेपी |
| (C) माना फलन $f:[-2, 2] \to (0, e^2]$ इस प्रकार की $f(x) = e^{\sqrt{x^2}}$, तो f(x) होगा | (R) विषम |
| (D) माना फलन $f: (-1, 5) \to [0, 3]$ इस प्रकार की $f(x) = \sqrt{5 + 4x - x^2}$, तो f(x) होगा | (S) अव्युत्क्रमणीय |
| | (T) आवर्तकालीय नहीं है |

**Ans.** (A) PRT (B) QRS (C) QST (D) QST

**Sol.** (A) We have $f(x) = \sqrt[3]{x} + \tan^{-1} x$

Clearly range of f(x) is R. (As f(x) is increasing in its domain.)

Hence f(x) is one-one and onto function, so f(x) is invertible.

Also f(x) is odd and aperiodic function.

(B) We have $f(x) = \sin^3 \pi\left(\text{sgn}(x^2 + 3x + 5)\right)$

(As $x^2 + 3x + 5 > 0 \ \forall \ x \in R$)

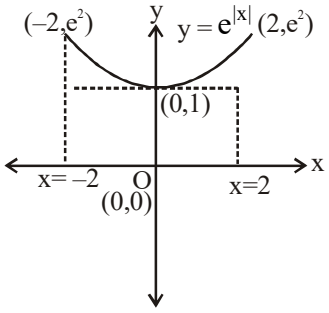
$\therefore$ $\text{sgn}(x^2 + 3x + 5) = 1$

Hence $f(x) = \sin^3 \pi = 0 \ \forall \ x \in R$

Clearly f(x) is into, odd and non-invertible.

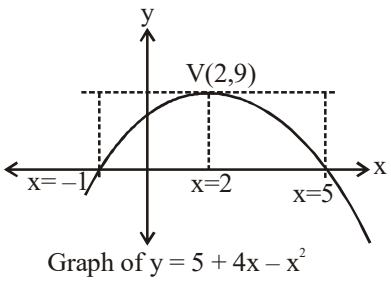
(C) We have $f(x) = e^{\sqrt{x^2}} = e^{|x|}$

Clearly f(x) is non-invertible and aperiodic and into function.




(D) We have $f(x) = \sqrt{5 + 4x - x^2}$

Clearly f(x) is into, non-invertible and aperiodic function.




Graph of $y = 5 + 4x - x^2$

19. Match the following column :

| **Column -I** | **Column -II** |
|---|---|
| (A) $f(x)=\sin^{-1}\left(\frac{2}{\lvert\sin x-1\rvert+\lvert\sin x+1\rvert}\right)$ | (P) f(x) is periodic |
| (B) $f(x)=\cos^{-1}\left(\lvert x-1\rvert-\lvert x-2\rvert\right)$ | (Q) Domain of f(x) is R |
| (C) $f(x)=\sin^{-1}\left(\frac{\pi}{\lvert\sin^{-1}x-\frac{\pi}{2}\rvert+\lvert\sin^{-1}x+\frac{\pi}{2}\rvert}\right)$ | (R) Range consists only irrational number |
| (D) $f(x)=\cos\left(\cos^{-1}\lvert x\rvert\right)+\sin^{-1}(\sin x)-\text{cosec}^{-1}(\text{cosec } x)+\text{cosec}^{-1}\lvert x\rvert$ | (S) f(x) is constant function |

निम्न कॉलम को सुमेलित कीजिए :

| **Column -I** | **Column -II** |
|---|---|
| (A) $f(x)=\sin^{-1}\left(\frac{2}{\lvert\sin x-1\rvert+\lvert\sin x+1\rvert}\right)$ | (P) f(x) आवर्ती है |
| (B) $f(x)=\cos^{-1}\left(\lvert x-1\rvert-\lvert x-2\rvert\right)$ | (Q) f(x) का प्रान्त R है |
| (C) $f(x)=\sin^{-1}\left(\frac{\pi}{\lvert\sin^{-1}x-\frac{\pi}{2}\rvert+\lvert\sin^{-1}x+\frac{\pi}{2}\rvert}\right)$ | (R) परिसर में अपरिमेय संख्याएं है |
| (D) $f(x)=\cos\left(\cos^{-1}\lvert x\rvert\right)+\sin^{-1}(\sin x)-\text{cosec}^{-1}(\text{cosec } x)+\text{cosec}^{-1}\lvert x\rvert$ | (S) f(x) अचर फलन है |

**Ans.** (A) PQRS (B) Q (C) RS (D) RS

**Sol.** (A) $f(x)=\sin^{-1}\left(\frac{2}{|\sin x-1|+|\sin x+1|}\right)$

$\because \sin x-1\le 0,\ \sin x+1\ge 0$

$f(x)=\sin^{-1}\left(\frac{2}{1-\sin x+1+\sin x}\right)$

$f(x)=\frac{\pi}{2}$

(B) $f(x)=\cos^{-1}\left(|x-1|-|x-2|\right)$

Case (I) $x\in(-\infty,1)$

$f(x) = \cos^{-1}(-x+1+x-2)$

$= \pi$

Case (II) $x \in (1,2)$

$f(x) = \cos^{-1}(-x+1-x+2)$

$= \cos^{-1}(3-2x)$

Case (III) $x \in (2,\infty)$

$f(x) = \cos^{-1}(x-1-x+2)$

$f(x) = 0$

(C) $f(x) = \sin^{-1}\left(\dfrac{\pi}{-\sin^{-1}x + \dfrac{\pi}{2} + \sin^{-1}x + \dfrac{\pi}{2}}\right)$

$= \dfrac{\pi}{2}$

(D) Domain = {–1, 1}

$f(\pm 1) = 1 + (\pm 1) - (\pm 1) + \dfrac{\pi}{2}$

Range $= 1 + \dfrac{\pi}{2}$